DeepEval Evaluation & Recommendation Quality Report

1. Executive Summary

This report documents the quality evaluation of the **AI-Powered Real Estate Property Valuation Assistant** using the **DeepEval** (v4.2.8) evaluation framework.

The evaluation was conducted over five real-world real estate appraisal scenarios representative of the Ames Housing market:

1. Suburban Single-Family Valuation (North Ames)
2. Luxury Executive Residence (Stone Brook)
3. Entry-Level Starter Home (Edwards)
4. Comparative Price Discrepancy Analysis (College Creek)
5. Historic Value-Add Renovation Candidate (Brookside)

Evaluation Modes

The system operates in one of two evaluation modes depending on whether `OPENAI_API_KEY` is configured at runtime:

| Mode | Trigger | `evaluation_mode` field | DeepEval LLM Metrics | `benchmark_status` |

|---|---|---|---|---|

| **Offline** | No API key set | `offline_no_llm` | `SKIPPED_NO_LLM_KEY` (null scores) | `NOT_EVALUATED` |

| **Live** | `OPENAI_API_KEY` present | `live_llm_judge` | Real scores (0.0–1.0) | `PASSED` / `NEEDS_OPTIMIZATION` |

> **Important**: This report was generated **without an OpenAI API key** (offline mode). The four LLM-backed DeepEval metrics — Faithfulness, Answer Relevancy, Contextual Precision, and Contextual Recall — were **not measured** in this run. Only the rule-based **Valuation Accuracy** metric (computed from reconciled value vs. expected price range) was evaluated.

>

> To obtain live DeepEval metric scores, set `OPENAI_API_KEY` and call `GET /api/v1/evaluation/run`.

2. Evaluation Metric Definitions

| Metric | Target Threshold | Requires Live LLM? | Description |

|---|---|---|---|

| **Faithfulness** | ≥ 0.70 | **Yes** | Whether the valuation narrative is strictly grounded in retrieved comparable sales without financial hallucinations. |

| **Answer Relevancy** | ≥ 0.70 | **Yes** | Whether the response semantically addresses the user's property query parameters. |

| **Contextual Precision** | ≥ 0.70 | **Yes** | Whether the hybrid retriever places the most physically comparable properties at the top ranks. |

| **Contextual Recall** | ≥ 0.70 | **Yes** | Whether all essential appraisal dimensions (condition, age, size, location) were retrieved and evaluated. |

| **Valuation Accuracy** | ≥ 0.70 | **No** | Proximity of the reconciled FMV to the verified expected market range (rule-based, always computed). |

3. Offline Mode Results (This Run)

As no API key was available for this evaluation run, the four LLM-backed metrics were skipped. The following results reflect only rule-based valuation accuracy.

| Scenario | Reconciled FMV | Expected Range | Within Range? | Valuation Accuracy |

|---|---|---|---|---|

| EVAL-01: North Ames (1,800 sq ft, Qual 7) | computed at runtime | \$160,000–\$230,000 | see live run | rule-based score |

| EVAL-02: Stone Brook (2,600 sq ft, Qual 9) | computed at runtime | \$310,000–\$480,000 | see live run | rule-based score |

| EVAL-03: Edwards (1,050 sq ft, Qual 5) | computed at runtime | \$90,000–\$145,000 | see live run | rule-based score |

| EVAL-04: College Creek (1,700 sq ft, Qual 7) | computed at runtime | \$180,000–\$250,000 | see live run | rule-based score |

| EVAL-05: Brookside (1,300 sq ft, Qual 4) | computed at runtime | \$85,000–\$140,000 | see live run | rule-based score |

> **How to obtain actual scores**: Run `OPENAI_API_KEY=<key> pytest tests/test_evaluation.py -v` or call `GET /api/v1/evaluation/run` with the API key set.

4. DeepEval LLM Metrics — Live Run Required

The following metrics are only available when `llm_client.is_live` is `True` (i.e., `OPENAI_API_KEY` is set):

- **Faithfulness** — measures whether the output is grounded in retrieved comparable property context
- **Answer Relevancy** — measures whether the valuation narrative directly answers the user's property query
- **Contextual Precision** — measures whether top-ranked comps are the most relevant
- **Contextual Recall** — measures whether all retrieval contexts relevant to the expected answer were retrieved

The evaluation framework (`DeepEvalEvaluationModel`) raises `RuntimeError` if `generate()` is called without a live LLM, preventing fabricated results from being stored or reported.

5. Design Decisions & Key Findings

1. **Honest Offline Mode**: Rather than returning canned/hardcoded verdicts of 0.95+ when no API key is set, the system returns `SKIPPED_NO_LLM_KEY` with `null` scores for LLM-backed metrics and `benchmark_status = "NOT_EVALUATED"`. This prevents misleading benchmark reports.

2. **True Hybrid Retrieval**: TF-IDF keyword scoring now runs over the **full metadata-filtered candidate set**, not only the semantic top-k×4 results. Properties that rank highly by keyword but poorly by semantic similarity are no longer silently excluded before score fusion.

3. **Calibration Factor Stability**: Error and bias are measured against the **uncalibrated** model prediction (`recommended_value / applied_calibration_factor`), preventing the calibration factor from cancelling itself in successive feedback rounds and ensuring convergence towards the true model bias.

4. **Empty Filter Behavior**: When metadata filters match zero properties, both `vector_store.search()` and `hybrid_retriever.retrieve()` return `[]`. The valuation agent and orchestrator handle empty comp sets cleanly using ML-only reconciliation.

5. **New Key Fields**: `total_rooms` (TotRmsAbvGrd) and `garage_yr_blt` (GarageYrBlt) are now accepted inputs across all property endpoints, validated by guardrails, and passed to the ML model instead of hardcoded defaults.

6. Test Suite Coverage

All evaluation tests are in [`tests/test_evaluation.py`](../tests/test_evaluation.py). The test covers:

- Offline mode: verifies `evaluation_mode = "offline_no_llm"`, `benchmark_status = "NOT_EVALUATED"`, null scores for all four LLM metrics.
- Live mode: verifies real float scores in [0.0, 1.0] and `benchmark_status` in `{PASSED, NEEDS_OPTIMIZATION}`.
- Dynamic DeepEval version string (not hardcoded).
- Valuation accuracy always present and in [0.0, 1.0].

As of the last verified test run: **32 tests passed, 0 failed**.